

1. Introduction

The **Fitness and Muscle Recovery System** is a mobile-first application designed to help users track physical training and physiological recovery. By utilizing an interactive anatomical map and structured workout logging, the system provides visual feedback on muscle fatigue and ensures consistent progress tracking through a mobile interface.

2. User Roles

- **Athlete/User:** The primary individual performing workouts, tracking fatigue via the muscle map, and managing exercise routines.
- **System Administrator:** Responsible for managing the global database of exercises and maintaining application updates.

3. Functional Requirements

1. Muscle Fatigue & Recovery Visualization

Users should be able to visualize their physical state through an interactive anatomical interface.

- **1.1** The system shall render a human body using **SVG paths** categorized into major muscle groups.
- **1.2** The system shall dynamically change the color of specific muscle paths based on **fatigue levels** (e.g., Red for high fatigue, Yellow for moderate, Green for recovered).
- **1.3** The system should implement a **recovery algorithm** to automatically transition muscle colors back to "recovered" states over a defined time period.
- **1.4** Users should be able to manually override or adjust the fatigue level of a muscle group if the automated estimation is inaccurate.

2. Exercise & Category Management

Users should be able to browse and select exercises to build their training sessions.

- **2.1** The system shall categorize exercises by target muscle group (e.g., Chest, Back, Legs).
- **2.2** The user shall be able to filter or search for specific exercises within the database.
- **2.3** The system should allow users to view details for each exercise, such as proper form or equipment needed.
- **2.4** The system shall categorize exercises by **Training Modality** (e.g., Weightlifting, CrossFit, Calisthenics, Track & Field, Yoga).

- **2.5** The system shall adapt the data entry fields based on the selected modality (e.g., Calisthenics tracks "Bodyweight + Added Load," Track & Field tracks "Distance & Time," while Weightlifting tracks "Sets, Reps & Weight").
- **2.6** The system should allow users to create and save "Custom Exercises" if a specific niche movement (e.g., a specific CrossFit WOD) is not in the global database.

3. Workout Execution & Tracking

Users should be able to record active training sessions in real-time.

- **3.1** The system shall provide a **"Start Workout"** interface to initiate a tracking session.
- **3.2** The system shall include a **timer/counter** to track the duration of the current workout or specific rest intervals.
- **3.3** Upon finishing a workout, the system shall automatically update the **fatigue status** of the involved muscle groups in the SVG map.
- **3.4** The system should allow users to input sets, reps, and weight used for each exercise during the session.
- **3.5** The system shall allow users to log the **RPE (Rate of Perceived Exertion)** on a scale of 1-10 for each set or completed workout.
- **3.6** The recovery algorithm shall factor in the user's logged RPE to adjust the fatigue severity on the SVG muscle map (e.g., an RPE 10 squat fatigues the system more than an RPE 7 squat, even if the weight is the same).
- **3.7** The system shall automatically calculate and display the **Total Volume Load** (Sets x Reps x Weight) for weightlifting sessions to help users track progressive overload.

4. Mobile Integration (with Capacitor)

The application must leverage native mobile capabilities for an enhanced user experience.

- **4.1** The interface must be **responsive**, ensuring the SVG muscle map is easily interactable on small touchscreens.
- **4.2** The system should utilize **Haptic Feedback** (vibration) to notify the user when a workout timer or rest period has expired.
- **4.3** The system shall support **Local Storage** to ensure user progress and muscle fatigue states are saved even if the app is closed.

5. AI Chat Assistant

Users should be able to interact with an intelligent assistant for guidance and motivation.

- **5.1** The system shall integrate an **AI Chatbot** to answer user queries regarding exercise form, nutrition, or recovery tips.
- **5.2** The AI assistant should be able to suggest **specific exercises** based on the current fatigue levels shown on the SVG muscle map.
- **5.3** The system should allow the AI to generate **personalized workout plans** based on the user's goals (e.g., hypertrophy, strength, or endurance).
- **5.4** The AI should provide **proactive encouragement** or "gentle warnings" if the user attempts to train a muscle group that is still in a high-fatigue (Red) state.

6. Multimodal Interaction & Accessibility

The app should support various ways to interact during a workout.

- **6.1** The user should be able to confirm the completion of a set via **Voice Command** (e.g., "Set finished") to allow for hands-free operation.
- **6.2** The system shall provide **Audio Feedback**, announcing the start of the next exercise or the end of a rest period.
- **6.3** The interface should adapt its **Visual Complexity** based on the user's profile, providing a simplified view for beginners or an advanced data-rich view for athletes.

7. History, Reporting & Monitoring

Users and trainers should be able to track long-term progress.

- **7.1** The system shall maintain a **detailed log** of all completed workouts, including timestamps and muscle fatigue snapshots.
- **7.2** The system should generate **Visual Reports** (charts and heatmaps) to illustrate training patterns over weeks or months.
- **7.3** Users should be able to **Export** their training and recovery data in standard formats like **PDF or CSV** to share with a personal trainer.
- **7.4** If the system detects a total lack of activity for a predefined duration, it should send a **Notification/Alert** to re-engage the user.

- **7.5** The system shall provide an **Interactive Calendar Interface** where days containing logged workouts are visually highlighted (e.g., using color-coded dots or rings based on the workout modality or intensity).
- **7.6** The system shall allow users to tap on any historical date within the calendar to view a detailed breakdown of that day's session (including exercises performed, volume load, and the logged RPE).
- **7.7** The calendar shall display a **"Fatigue Snapshot"** for past dates, allowing the user to see what their SVG Muscle Map looked like on that specific day.

8. Biometric & Environmental Integration

The app should adapt based on the user's physical state and surroundings.

- **8.1** The system shall sync with **Wearable Devices** to pull Heart Rate (HR) and Heart Rate Variability (HRV) to calculate "Readiness Scores".
- **8.2** The **SVG Muscle Map** should automatically adjust its recovery speed based on the user's sleep quality and protein intake logged in the system.
- **8.3** The system shall suggest **Optimal Training Environments** based on local weather (e.g., "It's 25°C and sunny; consider an outdoor cardio session")

9. Data Security & Privacy Management

The application must securely handle sensitive user health and biometric data.

- **9.1** The system shall securely encrypt user profile information and biometric data (like HRV and Weight) both in transit and at rest.
- **9.2** The system shall provide an explicit "Opt-In/Opt-Out" consent toggle for the AI Chat Assistant to use their personal fitness data for generating insights.
- **9.3** The system shall allow users to permanently delete their account and wipe all associated workout and physiological history from the database to comply with standard data privacy regulations (e.g., GDPR).

10. AI Visual Body Composition & Progress Tracking

The system shall utilize the device's camera and Computer Vision (CV) to analyze morphological changes in the user's body over time.

- **10.1** The system shall utilize a Computer Vision model to estimate **Body Fat Percentage** and **Lean Muscle Mass** by analyzing the user's bodily contours, logging these estimates in the user's progress history.
- **10.2** All images processed for body composition must be processed strictly locally on the device or heavily encrypted to ensure maximum user privacy.

4. Scenario: User Training Session

Context: Alex, a fitness enthusiast, is starting a "Leg Day" workout using the mobile app.

Flow:

1. Alex opens the app and views the **SVG Muscle Map**; his legs are currently Green (Recovered).
2. He navigates to the **Exercise Category** page and selects "Squats" and "Leg Press."
3. Alex taps **"Start Workout"**; a timer begins counting his session duration.
4. After finishing his last set, he hits "End Workout."
5. The system automatically updates his SVG map, changing the Quadriceps and Glute paths to **Red (High Fatigue)**.
6. Over the next 48 hours, the system gradually shifts these colors back to **Green** based on the recovery algorithm.

5. Scenario: AI-Guided Recovery

Context: Andreas, a software engineer, wants to train but feels sore from a previous session.

Flow:

1. Andreas opens the app on his **Smartwatch** and sees his Chest muscles are highlighted in **Yellow** (Moderate Fatigue).
2. He asks the **AI Assistant** via voice: "What should I train today?".
3. The AI analyzes his recovery data and replies: "Your chest is still recovering. I recommend a Leg-focused session today to optimize performance."
4. The system automatically loads a **Leg Routine** on his phone and wall-mounted display.
5. After the session, Andreas views a **Chart** on his phone showing his improved consistency for the month.

7. Scenario: The "Clever" Assistant

Context: Maria is a busy parent who only has 20 minutes to exercise.

Flow:

1. Maria opens the app; the **AI Assistant** notices her HRV is low, indicating high stress, and her "Sleep Score" was 60%.
2. Instead of her planned heavy lifting, the AI suggests: "Your recovery is low today. Let's do a 15-minute mobility flow to reduce cortisol".
3. She sets her phone on the floor; the **Camera** tracks her movements and confirms she is hitting the correct depth in her stretches.
4. The **Smartwatch** vibrates once she reaches her target heart rate zone.
5. The system logs the session and updates her **SVG Map** to a "Light Active Recovery" status.